

Dated: _____

Qno3. Pseudocode.

DECLARE Def : STRING

DECLARE Sum, marks, Average : Real

DECLARE N, student; INTEGER

INPUT N

// No. of students

FOR student = 1 to N

Sum = 0

FOR subj = 1 to 5

INPUT marks

Sum = Sum + marks

IF Mark < 33

THEN

Def = True

END IF

END FOR

Average = Sum / 5

IF Def == "True"

THEN

Print ("Fail Subject deficiency")

ELSE IF Average >= 80

THEN

Print ("Distinction")

ELSE IF Average >= 60

THEN

Print ("Pass")

ELSE

Print ("Fail")

END IF

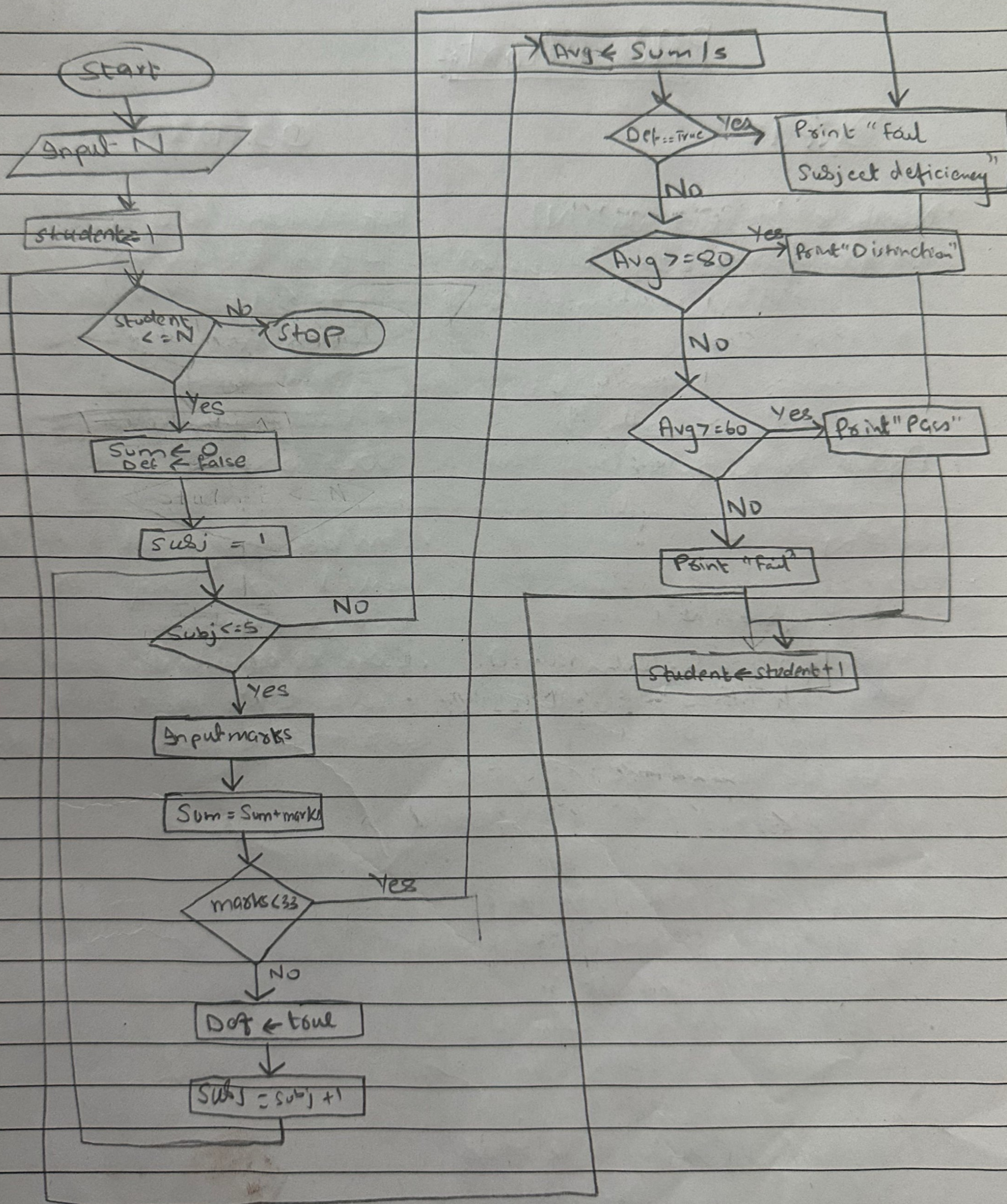
END IF

END IF

ENDFOR

Dated: _____

Qno 3:-



Dated: _____

PAC Chart:-

INPUT	OUTPUT
• Number of students (N) • Marks for 5 subjects (marks)	• For each student - student number - Average marks (Average) - Result category: "Distinction", "Pass", "Fail", or "Fail - Subject deficiency"
PROCESSING REQUIRED	SOLUTION ALTERNATIVES
• Read N • Loop for each student (1 to N) - Reset running sum (sum = 0) and deficiency flag (Def = "False") - Loop 5 times (1 to 5) • Input marks • Add marks to sum • If marks < 33, set Def = "True" - Compute Average = sum / 5 - If Def = "True", result is "Fail Subject deficiency" - Else if Average >= 80, result is "Distinction". - Else if Average >= 60, result is "Pass". - Else result is "Fail". - Print student's Average and Result	• Can be implemented using nested FOR loops or WHILE loops. • Subject deficiency override must take precedence over the calculated average.

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GPO chart

Input	Process	Output
• N (Total students) • marks (Mark for each subject, = per student)	1) Input 2) For each student (1 to N) a) initialise Sum = 0, Def = "True" b) for each Subj (1 to 5) i. input marks ii. Sum = Sum + marks iii. If marks < 33, Def = "True" c) calculate Average: Sum / 5. d) Determine Result • If Def == "True" → "Fail" • "Subject deficiency" • Else if Average > 80 → "Distinction" • Else if Average > 60 → "Pass" • Else → "Fail" e. Output student Summary	• Student number • Average • Result status